Chapter 5 Regression-Based Classification

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Regression-Based Classification

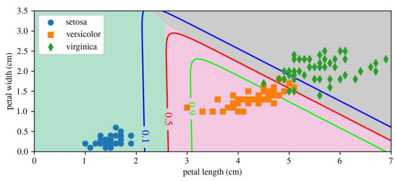
Regression models can often be adapted for classification.

Examples:

- Linear Regression → Logistic Regression
- Logistic Regression → Softmax Regression

Advantages:

- Interpretable models
- Familiar hyperparameters
- Probabilistic predictions



Classification regions for the three Iris species.

Logistic Regression

Logistic Regression estimates the probability that an input belongs to a class.

Predicted probability:

$$\hat{p} = h_\theta(X) = \sigma(\boldsymbol{\theta}^\top X)$$

Classification rule:

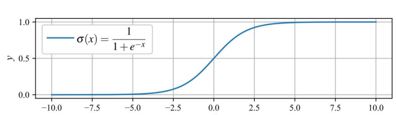
$$\hat{y} = \begin{cases} 0 & \hat{p} < 0.5 \\ 1 & \hat{p} \geq 0.5 \end{cases}$$

- Binary classifier
- Output is a probability
- Threshold typically set to 50%

The Sigmoid Function

The sigmoid function maps any real number to the interval $(0, 1)$: :contentReference[oaicite:0]index=0

- S-shaped curve
- Converts scores into probabilities
- Central component of Logistic Regression



Sigmoid function.

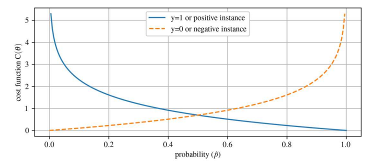
Classification Cost Function

Training Logistic Regression requires a cost function that:

- Rewards correct predictions
- Penalizes incorrect predictions
- Produces probabilities near:
  - 1 for positive examples
  - 0 for negative examples

Observations:

- Large penalty when a confident prediction is wrong
- Small penalty when a confident prediction is correct



Classification cost function.

Training Logistic Regression

Cost for a single sample:

$$C(\theta) = \begin{cases} -\log(\hat{p}) & y = 1 \\ -\log(1-\hat{p}) & y = 0 \end{cases}$$

Overall cost (log-loss):

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log(\hat{p}^{(i)}) + (1-y^{(i)})\log(1-\hat{p}^{(i)})\right]$$

- No closed-form solution
- Train using Gradient Descent
- Convex cost function
- Guaranteed global minimum

Data: Iris Flower Dataset

The Iris dataset contains measurements from:

- 150 flowers
- 3 species:
  - Iris-Setosa
  - Iris-Versicolor
  - Iris-Virginica

Features:

- Sepal length
- Sepal width
- Petal length
- Petal width

A classic classification dataset.

Three Iris flower species.

Data: Iris Measurements

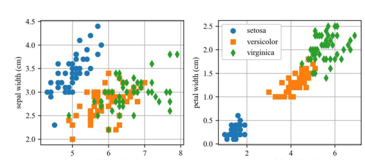
The four measured features can be used to classify flower species.

Observations:

- Setosa is easily separated
- Versicolor and Virginica overlap
- Petal measurements provide better separation than sepal measurements

Goal:

- Predict species from measurements



Sepal and petal measurements for the Iris dataset.

Example 5.1: Iris Dataset Exploration

- Load the Iris dataset from Scikit-Learn
- Examine feature and label information
- Visualize sepal and petal measurements
- Explore relationships between the three species

Demonstrates:

- Classification datasets
- Feature exploration
- Data visualization
- Class separation

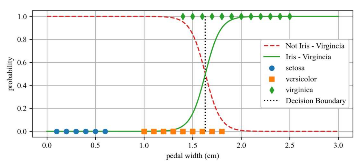
1-D Decision Boundaries

Logistic Regression can classify Iris-Virginica using a single feature:

Petal Width

Observations:

- Below 1 cm:
  - Confidently not Virginica
- Above 2 cm:
  - Confidently Virginica
- Between:
  - Model uncertainty



1-D decision boundary.

Decision Boundary

A Logistic Regression model predicts probabilities:

$$\hat{p} = P(y = 1|X)$$

where P denotes probability. In this case, $P(y = 1|X)$ means the probability that the sample belongs to class 1, given the input features X.

The classification boundary occurs when:

$$\hat{p} = 0.5$$

For the Iris dataset:

- Decision boundary occurs near:

$$\text{Petal Width} \approx 1.6 \text{ cm}$$

- Above the boundary: predict Iris-Virginica; below the boundary: predict not Iris-Virginica.

Example 5.2: 1D Logistic Regression

- Train a Logistic Regression model using:
  - Petal width only
- Estimate:
  - Probability of Iris-Virginica
- Visualize:
  - Predicted probabilities
  - 50% decision boundary
- Demonstrates how probabilities become class predictions

2-D Decision Boundaries

Using two features:

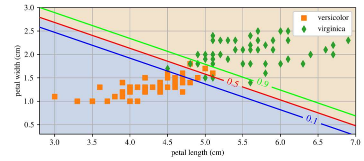
- Petal length
- Petal width

The model predicts:

$$P(\text{Virginica}|X)$$

Features:

- Dashed line:
  - 50% decision boundary
- Contours:
  - Equal probability levels
- Upper-right region:
  - High confidence Virginica



2-D decision boundary and probability contours.

Example 5.3: 2D Decision Boundary

- Train a Logistic Regression model using:
  - Petal length
  - Petal width
- Visualize:
  - Classification regions
  - Decision boundary
  - Probability contours
- Demonstrates how Logistic Regression separates classes in a two-dimensional feature space

Softmax Regression

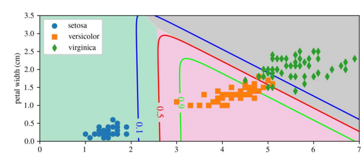
Softmax Regression extends Logistic Regression to multiple classes.

- No need to train several binary classifiers
- Works when each input belongs to exactly one class
- Also called Multinomial Logistic Regression

For an input vector $\mathbf{x}$, the model computes a score for each class.

Example:

- Classify all three Iris species simultaneously



Softmax classification of the three Iris species.

Class Scores and Probabilities

For class k, the score is

$$s_k(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\theta}^{(k)}$$

The softmax function converts scores into probabilities:

$$\hat{p}_k = \frac{\exp\left(s_k(\mathbf{x})\right)}{\sum_{j=1}^{K} \exp\left(s_j(\mathbf{x})\right)}$$

- $\mathbf{s}(\mathbf{x})$ is the vector of class scores
- K is the number of classes
- All predicted probabilities sum to 1

Prediction Rule

Softmax Regression predicts the class with the largest probability:

$$\hat{y} = \underset{k}{\operatorname{argmax}}\, \hat{p}_k = \underset{k}{\operatorname{argmax}}\, s_k(\mathbf{x}) = \underset{k}{\operatorname{argmax}} \left((\boldsymbol{\theta}^{(k)})^\top \mathbf{x} \right)$$

Note
The argmax operator returns the index of the largest value.

- Softmax Regression is multiclass, not multioutput
- It predicts one class at a time
- Useful for problems with mutually exclusive categories

Training Softmax Regression

The model is trained by minimizing the cross-entropy cost:

$$J(\Theta) = -\frac{1}{m} \sum_{i=1}^{m} \sum_{k=1}^{K} y_k^{(i)} \log \left(\hat{p}_k^{(i)} \right)$$

- Penalizes low probability for the true class
- When $K = 2$, this reduces to log loss
- No closed-form solution
- Train using Gradient Descent or another optimizer

Gradient with respect to class k:

$$\nabla_{\boldsymbol{\theta}^{(k)}} J(\Theta) = \frac{1}{m} \sum_{i=1}^{m} \left(\hat{p}_k^{(i)} - y_k^{(i)} \right) \mathbf{x}^{(i)}$$

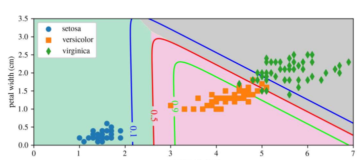
Example 5.4: Softmax Classification

The Iris dataset can be classified using petal length and petal width.

This example:

- Trains a multinomial logistic regression model
- Uses `multi_class="multinomial"`
- Requires a solver such as `lbfgs`

After training, a flower with 5 cm petals is classified as Iris-Virginica with probability 94.2%.



Softmax classification for the three Iris species.